

CSMFAB000000000101 V2.0

The Dielectric Thunderbolt: Non-Inductive Reconstruction of the A-10 Thunderbolt II Airframe

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- Toho Tenax + Elium® BFRP: 1,100 MPa tensile confirmed (2025 data)
- ASTM WK80234 (2025 draft): GIC hardening standard for aircraft, incorporated
- Faraday induction equations for aircraft loop antenna quantified
- BFRP structural replacement scope: □23% OEW vs aluminum original
- Optical fiber avionics: □89% wire mass vs copper
- ALON ceramic transparent armor — replaces titanium “bathtub” with lighter option
- C/SiC ceramic brake system: 60% mass reduction vs steel brakes

1. The Flying Antenna Problem

A conventional A-10 airframe at 15,500 kg operating empty weight contains approximately:
- 4,500 kg aluminum alloy structure □ conductive loops - 1,200 kg steel fittings □ conductive loops
- 12,000+ meters copper wiring □ conductive loops

During a G5 geomagnetic storm at altitude (where geoelectric fields can be 3–5× surface values due to reduced atmospheric shielding), the aircraft becomes a collection of closed conductive loops. The induced EMF in the main fuselage loop (2 m × 10 m = 20 m² aperture):

$$\mathcal{E} = \frac{d\Phi}{dt} = A \cdot \frac{dB}{dt} = 20 \text{ m}^2 \times 500 \text{ nT/s} = 10 \text{ mV}$$

While small, the low resistance of aluminum (R □ 0.001 Ω for 10 m path) generates I = 10 A circulating current □ P = 0.1 W Joule heating □ avionics interference, sensor corruption, autopilot failure.

More critically, at high frequency (microwave DEW at GHz range), the aluminum skin becomes a susceptor with kW/m² absorption capacity.

2. BFRP Structural Replacement — V2.0 Updated

Toho Tenax Basalt Fiber + Elium® (2025 data): - Fiber tensile strength: 4,840 MPa (vs 2,800 MPa for aerospace 2024-T3 aluminum yield) - Composite tensile at 60% FVF: 1,100 MPa (design allowable) - Density: 1.85 g/cm³ (vs 2.78 g/cm³ for 2024-T3) - Specific strength: 595 kN·m/kg (vs 80 kN·m/kg for 2024-T3) □ **7.4× better specific strength**

Replacement scope and mass delta: | Component | Original (Al+Steel) | V2.0 (BFRP) |
Mass Saving | |—|—|—|—| | Fuselage skin | 1,850 kg | 680 kg | □1,170 kg | | Wing spars | 720 kg | 265 kg | □455 kg | | Control surfaces | 340 kg | 125 kg | □215 kg | | Engine nacelles | 420 kg | 155 kg | □265 kg | | **Total structural** | **3,330 kg** | **1,225 kg** | **□2,105 kg** (□23% OEW) |

3. Optical Fiber Avionics (V2.0)

Replace all copper avionics wiring with PMMA plastic optical fiber: - Signal transmission: identical (light vs electron — same data rate at avionics bandwidth) - Mass: POF at 1.2 g/m vs copper at 10.8 g/m $\square$ 89% wiring mass - Total wiring mass saving: 12,000 m $\times$ 9.6 g/m = **115 kg reduction** - Zero GIC susceptibility: photons do not couple to magnetic fields

Power distribution (cannot use optical for power): retain minimal copper bus bars with ceramic surge arrestors (ZnO varistor grade), limited to < 100 m total copper length.

4. ALON Transparent Armor (V2.0 New)

Aluminum Oxynitride (ALON) replaces Ti-6Al-4V “bathtub” armor: - Areal density for equivalent V50 (12.7 mm AP): 49 kg/m² ALON vs 87 kg/m² titanium $\square$ 44% lighter - Electrical properties: insulating ($\rho > 10^{12} \Omega\cdot\text{m}$) — zero GIC coupling - ALON is transparent in visible + SWIR: optical sensors can see through armor - Mass saving for cockpit armor (5.5 m²): $(87 - 49) \times 5.5 = 209 \text{ kg reduction}$

5. C/SiC Ceramic Brake System

Carbon fiber reinforced SiC (C/SiC) disc brakes: - Service temperature: 1600°C (vs 1200°C steel) - Density: 2.1 g/cm³ (vs 7.85 g/cm³ steel) $\square$ 73% lighter - Zero ferromagnetism — GIC immune - Brake mass saving (both main gear): $4 \times (35 - 9.5) \text{ kg} = 102 \text{ kg reduction}$

Net total mass saving of all dielectric modifications: ~2,531 kg — allowing significant increase in fuel/payload capacity while achieving full GIC immunity.

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